



BSW

QUANT 101

2026 EDITION

Your Roadmap to Quantitative Finance

Quant 101 is a comprehensive roadmap built to guide students through the intricate recruitment landscape of top-tier quantitative finance. This document serves as a one-stop resource, consolidating the most relevant global contests, immersion programs, and application timelines from industry leaders like **Jane Street**, **Citadel**, and **Optiver**.

Beyond just a calendar, this guide integrates "**Senior Insights**", practical, first-hand advice on technical hurdles, mental math strategies, and interview logic. By bridging the gap between rigorous academic preparation and elite career opportunities, Quant 101 provides the clarity and strategic edge needed to excel in the world of high-frequency trading and quantitative research.



INSIGHTS



STRATEGY



RESOURCES



TIMELINES

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For detailed information, visit the full guide:
<https://docs.google.com/document/d/16XsGMZqThJnvy6DK-EYxYSmTorBrhq5d6FHfq35eWQ0/edit?usp=sharing>



PREPARATION RESOURCES

The Core Resource Stack - Quant Puzzles

Depth over Breadth: Pick one or two resources and finish it. Mastering the logic of 50 problems is infinitely more valuable than skimming 500.



ESSENTIAL RESOURCES



The Green Book - A Practical Guide to Quantitative Finance Interviews; starting point for probability, EV, and core quant fundamentals.



Brainstellar & PuzzledQuant - Build speed in puzzle-solving and recognize standard patterns.



QuantGuide.io - Highly recommended for building intuition across a wide range of problems.



50 Challenging Problems in Probability - F. Mosteller, critical for conditional probability, variance, and expectation.



LEVEL UP (AFTER GREEN BOOK)



Heard on the Street - Timothy Falcon Crack, blends quant questions with logic brain teasers from top hedge funds.



FAQ in Quantitative Finance - Paul Wilmott, brilliant breakdown of math, models, and industry practices for late-stage QR interviews.



Cut the Knot: Probability Riddles - Alexander Bogomolny, lateral thinking and unexpected problem-solving approaches for elite quant interviews.



Peter Winkler - Mathematical Puzzles, Advanced prep; many OA and interview questions are directly inspired by these.



CORE COURSE: MTL106 mastery is foundational for math rounds.



Master concepts. Solve intentionally. Think like a quant.



Platform Simulators for Mental Math

Theoretical knowledge is useless under extreme time pressure. Firms like **Optiver** and **Da Vinci** use intense mental math tests as primary filters. Drill daily.



TradeMath

The ultimate tool for exact OA replicas. Simulates the Optiver "80 in 8" numerical test, sequence tests, and the Da Vinci 16-minute quiz.

Key Features:



Optiver 80-in-8 replica



Da Vinci 16-min quiz



Dedicated market-making section



OpenQuant - Game Room

The best free platform for rapid-fire mental math. UI closely mirrors the actual assessments. Practice until consistently scoring 80+.

Key Modes:



Optiver 80-in-8



Task Switch



Memory Game



Target Score: Aim for a consistent 80+ on the Optiver 80-in-8 before applying. Speed AND accuracy under pressure are both evaluated.

★ BONUS

Finance Fundamentals & Bonus Resources

You don't need a finance background to crack quant interviews but having a conceptual overview of market mechanics gives you a **distinct edge** over purely tech-focused candidates.



Options, Futures & Other Derivatives

by John C. Hull

The industry "bible". Rigorous groundwork for option pricing models, including Black-Scholes, volatility smiles, and the Greeks.

Key Topics:



Black-Scholes Pricing Models



Volatility Smiles & Surfaces



The Greeks (Delta, Gamma, Vega)



Market Making Games

www.tradinginterview.com

The best free platform for rapid-fire mental math and market pricing intuition. UI closely mirrors actual assessments practice until consistently hitting 80+.

Key Modes:



Make Me A Market! – Facts & Guesstimates



Trading Interview Simulations



Derivatives Pricing Practice



Financial Derivatives & Pricing

<https://mellow-frown-cdc.notion.site/Financial-Derivatives-and-Pricing-Models-8026cc68ba254374b1dc4a2eee4eb018>



Probability Short Notes

<https://cloudy-lute-d02.notion.site/Probability-Primer-SoQ-2356c250015b4d17adfc6421653711f0>



Time Series Analysis Basics

<https://mellow-frown-cdc.notion.site/Time-Series-Analysis-194c61e63de949e69476cda92b258a87>



Pro Tip: You don't need all of these —pick one resource per gap. Consistent daily practice beats passive reading every time.



DSA & SYSTEMS

Data Structures, Algorithms & Core Systems

```
// DSA Operations
struct Node {
    int data;
    Node* next;
};
void insert() {
    // logic
}
int search() {
    // logic
}
```

```
// Graph Traversal
vector<int> bfs
(int s) {
    queue<int> q;
    // logic
}
// Data Structures
```



Competitive Programming

- Codeforces: target 1600–1700+ rating
- CSES problem set (mandatory)
- LeetCode / Striver for DSA-focused problems



C++ & OOP

- learncpp.com + cppreference.com
- Memory allocation & pointers
- OOP fundamentals for low-latency roles



Core CS & Systems

- OS: OSTEP; Arch: Patterson & Hennessy
- Networks: Kurose & Ross
- COL106 (DSA) is a baseline requirement



Machine Learning (for firms like BlackRock, TrexQuant, 26 Miles Capital)



Theory:

Andrew Ng ML Specialization / CS229 • StatQuest (YouTube)



Coding:

Hands-On ML (Géron) • CampusX 100 Days of ML • Know your metrics: F1, bagging, bootstrapping



General Interview Strategy & Tips

The technical bar is incredibly high, but interviewers primarily evaluate your **cognitive flexibility** and **how you react** when you don't immediately know the answer.



01

Think Out Loud

Communication is an evaluation metric. Articulate your thought process continuously. How you break down a problem is often more important than the perfect answer.



02

Embrace the Hints

Interviews are designed to push you until you're stuck. The key metric is your ability to catch, understand, and build upon hints. Treat it as a collaborative discussion.



03

Mental Math Speed

Speed and accuracy under pressure are vital in initial rounds. Practice fast mental math and estimation without relying on paper or calculators.



04

Group Discussions (GDs)

Quant GDs are often market-making games or strategy simulations. Be confident, proactive, and mathematically aggressive. Explain your strategy clearly.



05

CV Mastery

Be prepared to defend every line of your CV in depth. If you have technical projects (especially ML or systems), expect deep dives into architecture and the math.



06

Refer the Bluebook

For company-specific details, tips, and insider strategies on individual firms, always refer to the Bluebook for targeted preparation.



Programs at a Glance - 2026

A consolidated view of all major quantitative finance programs, contests, and immersion opportunities available to students in 2026. Deadlines are strict—mark your calendars.



PROGRAM		LOCATION / TYPE	EXPECTED DATES / DEADLINES (2026)
	IMC Trading Launchpad India	 Mumbai 2-Day Event	 May 18–19 or May 20–21, 2026
	Jane Street FTTP (India)	 India Multi-Day Event	 May 20–23, 2026 • Deadline: Mar 15, 2026
	Jane Street SEE	 Hong Kong Summer Program	 Typically July 2026 • Deadline: Early May 2026
	Optiver FutureFocus	 Mumbai 5-Day Immersive	 June 2026 • Deadline: Mid-April 2026
	Jump Trading Mumbai Internship	 Mumbai Winter Internship	 Dec 2026 – Feb 2027 • Bootcamp dates vary
	Tower Research 2-Day Program	 India Insight Program	 Typically held in May 2026
	Citadel Securities The India Terminal	 Virtual Coding Competition	 Deadline: Mar 8, 2026 • Program: Mar 23–30, 2026
	WorldQuant BRAIN IQC	 Global Team Competition	 Registration: Mar 16, 2026 • Stage 1: Mar 17 – May 18, 2026
	Five Rings LINK & Winter Internship	 India Dev Intensive + Internship	 Winter Internship: Jan 2027 • Applications: Feb/Mar 2026

 **Note:** Dates are subject to change. Always check official websites for the latest updates.



IMC TRADING

imc

IMC Trading | Launchpad India



Program Overview

A 2-day discovery event at the Mumbai office. Scheduled for **May 18–21, 2026** (two separate 2-day sessions).

Strictly requires a **CGPA of 8.0 or above**.

DAY 1



Cultural Immersion



General welcome, Mumbai office tour, interactive presentations on IMC's market-making philosophy.



You will be assigned a **dedicated mentor** to guide you through technical sessions.

DAY 2



The “Testing” Track



Trading Track: Market-making games and high-pressure trading simulations in real-time.



Technology Track: Algorithmic optimization challenges and technical coding exercises for low-latency execution.



Senior Insights



The “Links” Test

Before the Mumbai invite, expect a preliminary technical “Link” interview. For Tech track: C++ memory management and algorithmic efficiency.



Fast-Track Reward

Standout performers during these two days are **fast-tracked directly to final interview stages** for the next Summer Internship cycle.



Logic over Knowledge

They don't expect finance expertise they want to see how you **reason when a trading simulation goes against your position**.



Focus:
Orientation & Mentorship



Dates:
May 18–21, 2026



Location:
Mumbai Office

Jane Street | FTTP & SEE - “The Gold Standard”

FTTP (First-Year Trading and Technology Program) is a multi-day event in Hong Kong for first-year students.

SEE (Summer Exploratory Experience) is its counterpart for older students, typically held in July.

FTTP (India) is scheduled for **May 20–23, 2026**. Application deadlines generally fall in early May.



Trading Track Activities

1



Lectures

Short, intense sessions on probability, market structure, and game theory.

2



Mock Trading

Live simulations where you price assets and manage risk against peers.

3



The Estimation

A signature team event solving “Fermi problems” (e.g., “How many piano tuners are in Chicago?”) by providing logical confidence intervals.



Software Engineering Track

1



OCaml Immersion

Deep dive into functional programming using Jane Street’s primary language, OCaml.

2



System Design

Projects mimicking real-world exchange connectivity and data processing.



Fast-Track Reward

Successful completion of SEE or FTTP often allows you to **skip initial technical rounds** (like the “Puzzle or phone screens”) for a Summer Internship the following year.



The “Fermi” Filter

Practice estimating values within a logical **power-of-10 range** rather than seeking exact numbers.



The “No-Ego” Filter

Recruiters watch how you handle being wrong - acknowledge errors and **adapt immediately**.



Academic Excellence

Heavy emphasis on candidates with **IMO, IOI, or ICPC** competition backgrounds.



FTTP (India) Dates:

May 20–23, 2026



Application Deadline:

Early May



Location:

Hong Kong (SEE) | Mumbai (FTTP India)

Optiver | Future Focus & Ready Trader Go

Future Focus is a 5-day immersive program in Mumbai for 1st and 2nd-year students. **Ready Trader Go** is a global coding competition focused on building a market-making autotrader. The Mumbai program typically aligns with summer break in **June 2026**, with applications closing in **mid-April**.



Inside Future Focus - 5-Day Itinerary

01



Day in the Life

A full week at the Mumbai office learning how a global market maker operates.

02



Shadowing & Mentorship

Dedicated desk shadowing sessions – sit with traders and researchers to observe live market movements.

03



Trading Track

Interactive lectures on volatility, options theory, and risk, followed by simulated trading using Optiver's internal tools.

04



Tech Track

Hands-on workshops focused on building and optimizing low-latency systems for high-frequency execution.



Inside Ready Trader Go



The Challenge: Build and refine a market-making autotrader using a provided Python API.



The Simulation: Your bot competes on a simulated exchange against other teams, aiming to maximize profit while managing strict risk limits.



The Evaluation: Bots ranked on a leaderboard based on PnL and pricing strategy stability.



Senior Insights

- **The 80-in-8 Hurdle:** Optiver's famous 80 questions in 8 minutes mental math test. Aim for a consistent **75+ score** in practice to pass the initial screening link.
- **Coachability is King:** Optiver cares more about your learning curve than starting knowledge. Apply trader tips **immediately** in the next simulation.
- **Fast-Track Reward:** Both programs serve as a direct Pipeline standout participants are fast-tracked for **Summer 2027 internship offers**.



Future Focus Dates:
5 Days · June 2026



Applications Close:
Mid-April 2026



Location:
Mumbai, India

Jump Trading | Mumbai Internship & Bootcamp

Jump
TRADING

An intensive technical program focusing on low-level engineering.
Currently hiring for internships in Mumbai for the **Dec 2026 – Feb 2027** window.



The “Full-Cycle” Experience

Unlike some internships that focus on a single isolated project, Jump’s **10-week program** replicates the complete software development lifecycle, from initial design and coding to real production-ready deployment and testing.



Core Tech Stack

Highly distributed, low-latency trading systems using **C++** and **Python** (Pandas/NumPy for data analysis).



Training Phase

Intensive in-house courses on Market Mechanics, Trading Simulations, Machine Learning, and Linux Internals.



Project Work

Build and optimize high-performance data pipelines and develop tools directly supporting quantitative researchers.



Senior Insights



The Edge

Focus on **low-level systems programming**. Be prepared to answer questions on how software interacts with hardware: process scheduling, caching, and memory management.



Coding Style

They value **clean, maintainable, and highly efficient code** over clever hacks. “Thinking out loud” while solving non-typical, job-realistic coding problems is crucial.



The “No Bullsh*t” Policy

Interviewers are often engineers themselves who prefer **intellectual honesty**. If you don’t know an answer, explain your reasoning and admit where you are stuck rather than faking it.



Fast-Track Reward

The internship is the **primary pipeline for full-time hires** in Mumbai. Standout interns frequently receive return offers upon graduation.



Internship Window

Dec 2026 – Feb 2027



Program Duration

10 Weeks



Location

Mumbai, India



TOWER RESEARCH CAPITAL

Tower Research Capital | Insight & Hiring Challenges

Tower frequently hosts **2-day programs in May**. Access is often coordinated through the **OCS (Office of Career Services)** at **IIT Delhi**. Keep a close watch on the OCS portal the application window is often short.



Inside the 2-Day Program



01

Technical Sprints

Short, timed coding or research challenges that mimic the real-world pressure of high-frequency trading (HFT).



02

Market-Making Simulations

For the trading track – identify patterns in market data and adjust strategies in real-time.



03

System Design Deep Dives

For the developer track – low-latency optimization problems, focusing on how to shave **microseconds** off execution times.



04

Interaction with Senior Leads

Spend time with senior quantitative researchers and developers who provide feedback on your logic and coding style.



Senior Insights



Probability & Market Making Link

The preliminary screening is heavily weighted toward **Probability and Intuition-based Market Making**. Fast-paced mental math involving fractions and decimals is common.



C++ Mastery is Non-Negotiable

For the Dev track, Tower expects a very high level of C++ proficiency, specifically focusing on **performance optimization** and **memory management**.



The Gurugram Advantage

Since their main India office is in Gurugram, they frequently recruit from IIT Delhi. Watch the **OCS portal** closely.



Fast-Track Reward

Performing well in the 2-day insight program is a **direct path to an interview or a Summer 2027 internship offer**.



Program Duration

2 Days



Program Month

May



Recruitment Location

Gurugram, India



CITADEL SECURITIES

Citadel Securities | The India Terminal



CITADEL | Securities

A **virtual, week-long** competitive software engineering and game-theory challenge specifically for students in India. Teams of up to 3 build an algorithm to play a complex strategy game in a round-robin tournament against other student bots and a "boss bot" designed by Citadel engineers. You can use **Python, Java, or Rust**.



What Actually Happens



Virtual Gaming Environment

A games-based programming competition with a dynamic, real-time strategy environment.



The "Boss Bot" Challenge

Your team competes in a round-robin tournament against bots built by other students and Citadel engineers.



Real-Time Strategy

The game environment is dynamic – your bot must **adapt to opponents' moves in real-time**.



Senior Insights



Game Theory Over Finance

No finance background needed. Focus is entirely on **algorithmic efficiency and game theory**—Nash Equilibrium and expected value.



Team Collaboration

Citadel engineers observe how teams refine strategies over the week. They value a **"scientific approach"** – forming a hypothesis about why your bot lost and coding a logical fix.



The "Global Datathon" Parallel

If you enjoy this, also look into Citadel's **Datathons**, which focus more on messy datasets and predictive modeling.



Fast-Track Reward

Exceptional performance in The Terminal is a **massive signal on your CV**. Standout participants are often invited to interview for **Summer 2027** internships or full-time roles.



Format

Virtual, Week-Long



Team Size

Up to 3 Students



Languages

Python, Java, or Rust



Location

India (Virtual)

WorldQuant BRAIN | International Quant Championship (IQC)

A massive global team-based competition (1–4 members) where you build “alphas” (predictive mathematical signals) using WorldQuant’s web-based simulation platform.



Registration Opens: **March 16, 2026** | Stage 1 (Qualifier): **March 17 – May 18, 2026**

Inside the Competition

01



Stage 1: The Alpha Build

Over two months, teams use WorldQuant’s datasets (price, volume, and fundamental data) to create predictive signals.

02



Learning Phase

WorldQuant provides “Learn2Quant” resources including live webinars on Fundamental & Model Data, Price Volume Data, and tips for building “submittable Alphas”.

03



The Platform

Work in a sandbox environment testing mathematical formulas against historical market data to see their **Sharpe Ratio** and **Drawdown**.

04



Advancement

Top scorers from Stage 1 move to the National Finals, eventually leading to a **Global FINAL** event.



Senior Insights



Quantity AND Quality

The BRAIN scoring system rewards **consistency**. Don’t just aim for one “perfect” alpha – build a library of high-performing, unique signals to maximize your score.



No Finance Background Needed

Many top performers are students in **Math, CS, or Physics** who treat the market as a pure signal-to-noise problem.



The “Consultant” Pipeline

High-scoring participants are often invited to become **WorldQuant BRAIN Research Consultants** – a paid, flexible role that often leads to full-time Quantitative Researcher offers.



Team Synergy

Form a team with **diverse skills** – one person on data analysis, another on mathematical formulas, and one on optimizing “alpha” performance.



Team Size

1 – 4 Members



Format

Global, Virtual



Platform

WorldQuant (Web)



Outcome

Prizes + Consultant Offers

Five Rings | LINK & Winter Internship

LINK is a Software Development Intensive program. The **Winter Internship (January 2027)** is a high-intensity, 4-week immersion in mock trading and strategy games. Applications for Winter 2027 opened on **February 24, 2026**.



Inside the Program



Classroom to Desk Transition

The first week is dedicated to intensive classroom training on **market mechanics** and **trading strategy**.



Real-World Impact

Interns develop and test **actual trading system components** that can be integrated into their live environment—no toy projects.



Mock Trading & Strategy Games

A core part of the experience involves playing proprietary strategy games designed by Five Rings traders to test **decision-making under uncertainty**.



Collaborative Engineering

For LINK and dev-focused roles, work closely with senior engineers to translate complex mathematical models into **high-performance C++ code on Linux**.



Senior Insights



The “Mental Math” Wall

Five Rings has one of the most aggressive mental math and probability screening processes in the industry. Practice until you can calculate **expected values for complex bets in seconds**.



The “Book of Five Rings” Philosophy

Named after Miyamoto Musashi’s classic, the firm values “**seeing what is not obvious**”. In interviews, they look for candidates who can find the hidden edge in a simple-looking game.



Tech Stack Precision

If applying for LINK or Dev track, your **C++ proficiency must be top-tier**, specifically regarding how code interacts with the Linux kernel for low-latency execution.



Fast-Track Reward

The Winter Internship is a primary “**trial run**” for full-time roles. Standout interns frequently receive **full-time return offers** by the end of the four-week period.



Quant 101 is an initiative by the **IIT Delhi BSW Board**.

For queries, contact Aniket Anand (9304497306), Aditya Singal (9331045664), Aryan Rana (8373939236)